

Step 1: calibration samples

$$\mathcal{C} = \{a_i = (x_i, y_i, z_i)^\top\}_{i=1}^n$$

Step 2: gravity estimate

$$g_0 = \frac{1}{n} \sum_{i=1}^n a_i \quad s = \|g_0\|_2 \quad \hat{d} = \frac{g_0}{s}$$

Step 3: horizontal basis

$$u_x = \frac{e_x - (e_x^\top \hat{d}) \hat{d}}{\|e_x - (e_x^\top \hat{d}) \hat{d}\|_2}$$

$$\|e_x - (e_x^\top \hat{d}) \hat{d}\|_2 \approx 0 \Rightarrow u_x = \frac{e_y - (e_y^\top \hat{d}) \hat{d}}{\|e_y - (e_y^\top \hat{d}) \hat{d}\|_2}$$

$$u_y = \frac{\hat{d} \times u_x}{\|\hat{d} \times u_x\|_2}$$

Step 4: normalized outputs

$$a_x^{(g)}(t) = \frac{a(t)^\top u_x}{s} \quad a_z^{(g)}(t) = \frac{a(t)^\top \hat{d}}{s} - 1$$

Variables

$a_i, a(t)$	$= (x, y, z)^\top$	$\mathcal{C}$	$= \{a_i\}_{i=1}^n$
n	$=$ calibration sample count	g_0	$=$ calibration mean vector
s	$=$ counts per g	$\hat{d}$	$=$ unit down vector
e_x	$= (1, 0, 0)^\top$	e_y	$= (0, 1, 0)^\top$
u_x	$=$ horizontal x unit vector	u_y	$=$ horizontal y unit vector
$a_x^{(g)}(t)$	$=$ x acceleration in g	$a_z^{(g)}(t)$	$=$ z acceleration in g